ON EQUIVARIANT PREFACTORIZATION ALEBRAS

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Introduction

Given a topological manifold M, we can construct an ∞-operad $\mathbb{E}_M$ of "little disks" in M. A fundamental result in the theory of prefactorization algebras is that the homotopy theory of $\mathbb{E}_M$-algebras admits an alternative, more rigid description in terms of *locally constant disk algebras* [Lur17, Theorem 5.4.5.9]. More precisely, there is a (discrete) operad $\mathrm{Disk}(M)$ such that, for every symmetric monoidal ∞-category $\mathcal{V}$, we have an equivalence

$$\mathrm{Alg}^{\mathrm{lc}}_{\mathrm{Disk}(M)}(\mathcal{V}) \simeq \mathrm{Alg}_{\mathbb{E}_M}(\mathcal{V}).$$

Here "lc" means taking the full subcategory of algebras that carry each unary operation to an equivalence.

In many cases, M carries a natural action of a topological group G, and we would like to think of an equivariant version of the above story. Essentially by definition, $\mathbb{E}_M$ admits a G-action, so what we want is a tractable operad $\mathrm{Disk}_G(M)$ that admits an equivalence

$$\mathrm{Alg}^{\mathrm{lc}}_{\mathrm{Disk}_G(M)}(\mathcal{V}) \simeq \mathrm{Alg}_{\mathbb{E}_M}(\mathcal{V})^G.$$

The difficulty in obtaining an operad $\mathrm{Disk}_G(M)$ is that $\mathrm{Disk}(M)$, being a discrete operad, does not admit an interesting action by G. The goal of this note is to show that this is nevertheless possible.

To define $\mathrm{Disk}_G(M)$, we observe that if we forget about the topology of G, then it acts on $\mathrm{Disk}(M)$. We define $\mathrm{Disk}_G(M)$ by forming the Grothendieck construction of this action, and then putting what looks like a natural topology on it.

Definition 0.1. Let G be a topological group, and let M be a d-manifold with a left G-action. We define a topological operad $\mathrm{Disk}_G(M)$ as follows:

(1) Objects are the open sets $U \subset M$ that are homeomorphic to $\mathbb{R}^d$.
(2) The mapping space $\mathrm{Disk}_G(M)\binom{(U_i)_{i\in I}}{V}$ is the subspace of G^I spanned by the elements $(g_i)_{i\in I}$ such that the open sets $(g_iU_i)_{i\in I}$ are disjoint and are contained in V.

We write $|\mathrm{Disk}_G(M)|$ for the ∞-operadic localization of $\mathrm{Disk}_G(M)$ at all unary morphisms.

The definition of $\mathrm{Disk}_G(M)$ looks too naïve to encode "up to homotopy" information. For example, it is not clear whether a homotopy equivalence $G' \xrightarrow{\simeq} G$ of topological groups induces an equivalence $|\mathrm{Disk}_{G'}(M)| \xrightarrow{\simeq} |\mathrm{Disk}_G(M)|$. However, it turns out that this is actually the correct definition:

Theorem 0.2. *Let M be a topological manifold with a left action of a topological group G. There is an equivalence of ∞-operads*

$$|\mathrm{Disk}_G(M)| \simeq \mathrm{colim}_{BG}\,\mathbb{E}_M.$$

The proof of Theorem 0.2 is interesting in its own right: Essentially, what we will do is to decompose the colimit on the right-hand side into pieces, and then glue them back in a different way to get the left-hand side. What lies at the heart of

the decomposition process is the "flattening construction" of Dwyer–Kan, which we review in Section 2. The decomposition will then be carried out in Section 3. We will then analyze the pieces carefully Section4, and then glue them back in Section 5.

Notation and terminology.

Convention 0.3. All topological categories or topological operads are regarded as simplicial operads by taking the singular complex of the mapping spaces.

Convention 0.4. Unless there is a chance of confusion, we generally do not distinguish between ordinary categories and their nerves. A similar remark applies to operads.

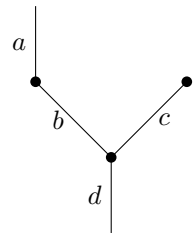
Definition 0.5. Given a topological group G, we write BG for the topological with a single object, denoted by $*$, whose endo-mapping space is given by G. By abusing notation, we will use the same notation BG to denote the homotopy coherent nerve of this enriched category.

Definition 0.6. Given an ∞-operad $\mathcal{O}$, we write $|\mathcal{O}|$ for its localization at all unary morphisms.

Warning 0.7. The "invert all unary maps" functor

$$|-| : \mathcal{O}\mathsf{p}_\infty \to \mathcal{O}\mathsf{p}_\infty$$

is *not* a left adjoint, because it does not preserve small colimits. To see this, consider a tree T depicted in the following picture:



There are two non-identity unary operations in (the free operad generated by) T: The maps $a \to b$ and $a \to d$. Inverting all these maps, one gets an operad whose underlying category has at most (in fact, exactly) two components. On the other hand, we can decompose T as $T = T_1 \amalg_b T_2$, where T_1 consists of b, c, d and the vertices attached to c, and T_2 consists of a, b and the vertex between them. The pushout $|T_1| \amalg_{|b|} |T_2|$ can be identified with T_1, so its underlying category has three components. Hence $|T|$ is not equivalent to $|T_1| \amalg_{|b|} |T_2|$.

Remark 0.8. Despite Warning 0.7, the next best thing is true: For any diagram $\mathcal{O}_\bullet : \mathcal{I} \to \mathcal{O}\mathsf{p}_\infty$, the map

$$|\mathrm{colim}_{i \in \mathcal{O}} \mathcal{O}_i| \to |\mathrm{colim}_{i \in \mathcal{O}} |\mathcal{O}_i||$$

is always an equivalence. (The proof of this uses the standard result on epimorphism in ∞-categories, applied to $\mathcal{O}_\bullet \Rightarrow |\mathcal{O}_\bullet|$; see [Ram22, Proposition A.11].)

1. Review on $\mathbb{E}_M$-operad

In this section, we recall the precise definitions of the $\mathbb{E}_M$-operad and associated constructions and theorems.

Definition 1.1. Let $d \geq 0$. We will write $\mathcal{M}\mathsf{fld}\,(d)^{\Delta}$ for the simplicial category of d-manifolds; the mapping spaces given by the singular complexes of embedding

spaces with the compact-open topology. We will write $B\,\mathrm{Emb}\,(d)^{\mathbf{\Delta}} \subset \mathcal{M}\mathsf{fld}\,(d)^{\mathbf{\Delta}}$ for the full simplicial subcategory spanned by $\mathbb{R}^d$.

We let $\mathcal{M}\mathsf{fld}\,(d)$ and $B\,\mathrm{Emb}\,(d)$ denote the homotopy coherent nerve of $\mathcal{M}\mathsf{fld}\,(d)$ and $B\,\mathrm{Emb}\,(d)^{\mathbf{\Delta}}$.

Remark 1.2. By Kister–Mazur's theorem [Kis64], the inclusion $B\,\mathrm{Top}\,(d) \to B\,\mathrm{Emb}\,(d)$ is an equivalence of ∞-categories. In particular, $B\,\mathrm{Emb}\,(d)$ is an ∞-groupoid, justifying its notation.

Definition 1.3. We define the **tangent classifier** functor

$$\tau : \mathcal{M}\mathsf{fld}\,(d)^{\mathbf{\Delta}} \to \mathsf{sSet}_{/B\,\mathrm{Emb}(\mathbb{R}^d)}$$

by $\tau\,(M) = B\,\mathrm{Emb}\,(d)_{/M}$.

Remark 1.4. We refer to Remark 2.8 for a detailed description of the simplicial enrichment of τ. Also, see [Ara25, Remark 2.6] for the explanation on the terminology "tangent classifier."

Definition 1.5. [Lur17, Definition 5.4.2.10] There is a simplicial functor

$$\mathbb{E}_\bullet \colon \mathsf{sSet}_{/B\,\mathrm{Emb}(d)} \to \mathsf{sSet}_{/\mathsf{Fin}_*},$$

which, when applied to a Kan fibration $p\colon B \to B\,\mathrm{Emb}\,(d)$, produces an ∞-operad $\mathbb{E}_B$. Explicitly, $\mathbb{E}_B$ is the suboperad of $B\,\mathrm{Emb}\,(d)^{\amalg}$ consisting of those operations $(\alpha_i)_{i\in I} : (b_i)_{i\in I} \to b$ such that the images of the maps $p\,(\alpha_i) : \mathbb{R}^d \to \mathbb{R}^d$ are mutually disjoint.

If M is a d-manifold, we write $\mathbb{E}_M = \mathbb{E}_{\tau(M)}$.

Remark 1.6. Since $\mathbb{E}_\bullet$ is a functor from $\mathcal{M}\mathsf{fld}\,(d)$ to $\mathcal{O}\mathsf{p}_\infty$, if M is a d-manifold with a left action of a topological group G, then $\mathbb{E}_M$ can be enhanced to a functor

$$\mathbb{E}_M \colon BG \to \mathcal{O}\mathsf{p}_\infty.$$

Next, we recall the discrete version of $\mathbb{E}_M$:

Definition 1.7. Let M be a d-manifold. We define a colored operad $\mathrm{Disk}\,(M)$ as follows:

(1) Its objects are the open subsets of M that are homeomorphic to $\mathbb{R}^d$.
(2) The set $\mathrm{Disk}\,(M)\left(\begin{smallmatrix}(U_i)_{i\in I}\\ V\end{smallmatrix}\right)$ is either a singleton or empty; it is a singleton if and only if the U_i's are disjoint and contained in V.

The following theorem is a fundamental result in the theory of prefactorization algebras.

Theorem 1.8. [Lur17, Theorem 5.4.5.9] *The map*

$$|\mathrm{Disk}\,(M)| \to \mathbb{E}_M$$

is an equivalence of ∞-operads.

2. Flattening and thickening

This section will focus on Dwyer–Kan's *flattening construction* [DK85, §7] and the closely related *thickening construction*. We will recall that they provide a particularly nice way to *de*-localize simplicial categories into ordinary categories (Proposition 2.3). We also discuss the compatibility of these constructions with slice ∞-categories (Proposition 2.7). The results in this section will be used later to "decompose" the G-action on $\mathbb{E}_M$.

We first start by recalling the flattening and thickening constructions.

Definition 2.1. Let $\mathcal{C}$ be a simplicial category. For each $n \geq 0$, let $\mathcal{C}^{\Delta^n}$ denote the simplicial category with the same objects as $\mathcal{C}$, with mapping simplicial sets given by

$$\mathcal{C}^{\Delta^n}(x,y) = \mathrm{Map}(\Delta^n, \mathcal{C}(x,y)).$$

We write $\mathcal{C}_n$ for the underlying category of $\mathcal{C}^{\Delta^n}$.

The **thickening** $\mathrm{Thick}(\mathcal{C})$ of $\mathcal{C}$ is the Grothendieck construction $\int \mathcal{C}^{\Delta^\bullet} \to \mathbf{\Delta}^{\mathrm{op}}$ of $\mathcal{C}^{\Delta^\bullet} \colon \mathbf{\Delta}^{\mathrm{op}} \to \mathsf{Cat}_\Delta$. Thus, it is a simplicial category whose objects are pairs $([n], x)$ for $n \geq 0$ and $x \in \mathcal{C}$, and whose mapping simplicial sets are given by

$$\left(\int \mathcal{C}^{\Delta^\bullet}\right)(([n],x),([m],y)) = \mathrm{Hom}_{\mathbf{\Delta}}([m],[n]) \times \mathcal{C}^{\Delta^m}(x,y).$$

The **flattening** $\mathrm{Flat}(\mathcal{C})$ of $\mathcal{C}$ is the Grothendieck construction $\int \mathcal{C}_\bullet \to \mathbf{\Delta}^{\mathrm{op}}$ of the functor $\mathcal{C}_\bullet \colon \mathbf{\Delta}^{\mathrm{op}} \to \mathsf{Cat}$. Equivalently, it is the underlying category of $\mathrm{Thick}(\mathcal{C})$.

Remark 2.2. We can define the thickening and flattening of simplicial operads, too, using the operadic Grothendieck construction.

The utility of the flattening construction comes from the fact that it provides a delocalization of simplicial categories:

Proposition 2.3. *Let $\mathcal{C}$ be a locally Kan simplicial category. The simplicial functors*

$$\mathrm{Flat}(\mathcal{C}) \hookrightarrow \mathrm{Thick}(\mathcal{C}) \hookleftarrow \mathcal{C}^{\Delta^0} \cong \mathcal{C}$$

induce equivalences of ∞-categories

$$\mathrm{N}(\mathrm{Flat}(\mathcal{C}))[\mathrm{cc}^{-1}] \underset{\phi}{\xrightarrow{\simeq}} \mathrm{N}_{\mathrm{hc}}(\mathrm{Thick}(\mathcal{C}))[\mathrm{cc}^{-1}] \underset{\psi}{\xleftarrow{\simeq}} \mathrm{N}_{\mathrm{hc}}(\mathcal{C}),$$

where cc *denotes the set of cocartesian morphisms over $\mathbf{\Delta}^{\mathrm{op}}$.*

A similar claim holds for locally Kan simplicial operads.

Proposition 2.3 appears as [DK87, 2.5] (where they proved a stronger claim for non-locally-Kan simplicial categories). We give an alternative, perhaps more conceptual proof of it below. Before the proof, let us record the following corollary of Proposition 2.3:

Corollary 2.4. *Let $\mathcal{C}$ be a locally Kan simplicial category. We have a natural equivalence*

$$\mathrm{N}_{\mathrm{hc}}(\mathcal{C}) \simeq \mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathrm{N}(\mathcal{C}_n)$$

in $\mathcal{C}\mathsf{at}_\infty$. A similar claim holds for locally Kan simplicial operads.

Proof. This follows from the fact that localizing the Grothendieck construction at cocartesian morphisms computes colimits in $\mathcal{C}\mathsf{at}_\infty$ [Lur09, Corollary 3.3.4.3]. □

We now turn to the proof of Proposition 2.3. We need a few lemmas.

Lemma 2.5. *For any simplicial set X, the map*

$$\mathrm{hocolim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} X_n \to \mathrm{hocolim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathrm{Map}(\Delta^n, X)$$

is a weak homotopy equivalence.

Proof. Homotopy colimits of simplicial objects in sSet are computed by taking the diagonal of the associated bisimplicial set. Thus, our task is to show that the map

$$X \to \widetilde{X}$$

is a weak homotopy equivalence, where $\widetilde{X} \in \mathsf{sSet}$ is the simplicial set whose n-simplices are the maps $\Delta^n \times \Delta^n \to X$. We can identify this map with the map

$$\mathrm{hocolim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} X \to \mathrm{hocolim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathrm{Map}(\Delta^n, X),$$

which is a weak homotopy equivalence because the map $X \to \mathrm{Map}(\Delta^n, X)$ is one for every $n \geq 0$. □

For the next lemma, we will write $\mathsf{Cat}_{\Delta,S}$ for the category of simplicial categories with a fixed set of object S, and simplicial functors that induce identity maps on S. Recall that this category admits a model structure transferred from $\mathsf{sSet}^{S\times S}$ [Lur17, Proposition 4.1.8.3].

Lemma 2.6. *Let S be a set. The forgetful functor*

$$U\colon \mathsf{Cat}_{\mathbf{\Delta},S} \to \mathsf{Cat}_{\mathbf{\Delta}}$$

preserves homotopy colimits indexed by weakly contractible categories.

Proof. The functor U factors as

$$\mathsf{Cat}_{\mathbf{\Delta},S} \xrightarrow{F} (\mathsf{Cat}_{\mathbf{\Delta}})_{S/} \xrightarrow{G} \mathsf{Cat}_{\mathbf{\Delta}}.$$

We will show that F and G preserves homotopy colimits.

For F, it suffices to show that it is left Quillen. To see this, we observe that F has a right adjoint. Explicitly, it carries an object $f\colon S \to \mathcal{C}$ to the simplicial category $\mathcal{C}_S \in \mathsf{Cat}_{\mathbf{\Delta},S}$ with $\mathcal{C}_S(x,y) = \mathcal{C}(f(x), f(y))$. This right adjoint is right Quillen, so F is left Quillen, as claimed.

The claim for G follows from [Lur09, Proposition 4.4.2.9] and [Cis19a, Corollary 7.6.13]. □

Proof of Proposition 2.3. We first consider the case of simplicial categories. According to [Lur09, Corollary 3.3.4.3], we can identify ϕ and ψ with the maps

$$\mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathrm{N}(\mathcal{C}_n) \xrightarrow{\phi'} \mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}^{\Delta^n}\right) \xleftarrow{\psi'} \mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}^{\Delta^0}\right)$$

in $\mathcal{C}\mathsf{at}_\infty$. The map ψ' is an equivalence because $\mathbf{\Delta}^{\mathrm{op}}$ is weakly contractible and the map $\mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}^{\Delta^0}\right) \to \mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}^{\Delta^n}\right)$ is an equivalence. To show that ϕ' is an equivalence, we use Lemma 2.6: It suffices to show that the map

$$\mathrm{hocolim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathcal{C}_n \to \mathrm{hocolim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathcal{C}^{\Delta^n}$$

is a weak equivalence of $\mathsf{Cat}_{\Delta,\mathrm{ob}\,\mathcal{C}}$. For this, notice that the functor $\mathsf{Cat}_{\Delta,\mathrm{ob}\,\mathcal{C}} \to \mathsf{sSet}^{S\times S}$ preserves sifted colimits [Lur17, Lemma 4.1.8.13, Proposition 3.2.3.1], and that it preserves and detects weak equivalences. Thus, it suffices to show that, for each pair of objects $x, y \in \mathcal{C}$, the map

$$\mathrm{hocolim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathcal{C}_n(x,y) \to \mathrm{hocolim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathcal{C}^{\Delta^n}(x,y)$$

is a weak homotopy equivalence. This follows from Lemma 2.5.

The above proof works almost verbatim in the case of simplicial operads. The only difference is that we need to replace $\mathsf{Cat}_{\mathbf{\Delta}}$ by $\mathsf{Op}_{\mathbf{\Delta}}$ and $\mathsf{sSet}^{S\times S}$ by $\mathsf{sSet}^{\coprod_{n\geq 0} S^n\times S}$. The relevant model structure for $\mathsf{Op}_{\mathbf{\Delta}}$ and $\mathsf{Op}_{\mathbf{\Delta},S}$ are discussed in [HM22, Example 13.32] and [HM22, Theorem 14.12], and the preservation of sifted homotopy colimits for $\mathsf{Op}_{\mathbf{\Delta},S} \to \mathsf{sSet}^{\coprod_{n\geq 0} S^n\times S}$ is discussed in [WY24, Proposition 7.2.1]. □

We end this section by recording the compatibility of the slice construction with the flattening construction. The proof requires a somewhat delicate care of combinatorics, and we recommend readers to either take the result as a black box or come back to it when the need arises.

Proposition 2.7. *Let $\mathcal{C}$ be a locally Kan simplicial category. Consider the simplicial functor*

$$\begin{aligned} \mathrm{N}\left(\mathrm{Flat}\left(\mathcal{C}\right)\right) &\to \mathcal{S}, \\ \left([n], x\right) &\mapsto \mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}^{\Delta^n}\right)_{/x}. \end{aligned}$$

This functor sends cocartesian morphisms to equivalences, and the induced functor

$$\mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}\right) \simeq \mathrm{N}\left(\mathrm{Flat}\left(\mathcal{C}\right)\right)\left[\mathrm{cc}^{-1}\right] \to \mathcal{S}$$

is naturally equivalent to the functor $x \mapsto \mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}\right)_{/x}$.

Remark 2.8. In the situation of Proposition 2.7, it is not entirely clear how the assignment $x \mapsto \mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}\right)_{/x}$ is made into a functor $\mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}\right) \to \mathcal{S}$. A closer inspection of the proof of Proposition 2.7 shows that this coincides with the nerve of the composite

$$\mathcal{C} \xrightarrow{\text{enriched yoneda}} \mathrm{Fun}^{\mathbf{\Delta}}\left(\mathcal{C}, \mathsf{sSet}\right) \xrightarrow{\mathrm{Un}} \mathsf{sSet}_{/\mathcal{C}} \xrightarrow{\text{forget}} \mathsf{sSet},$$

where Un denotes the unstraightening functor of [Lur09, §2.2.1]. (We refer to [Ara24b, Remark 4.6] for the simplicial enrichment of Un.)

Proof of Proposition 2.7. The claim on cocartesian morphisms is obvious, so it will suffice to extend the functor

$$\mathrm{Flat}\left(\mathcal{C}\right) \to \mathsf{sSet},\ \left([n], x\right) \mapsto \mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}^{\Delta^n}\right)_{/x}$$

to a simplicial functor $\mathrm{Thick}\left(\mathcal{C}\right) \to \mathsf{sSet}$. To this end, let $m, n \geq 0$ and $x, y \in \mathcal{C}$. We define a map

$$\mathrm{Hom}_{\mathbf{\Delta}}\left([m], [n]\right) \times \mathcal{C}^{\Delta^m}\left(x, y\right) \to \mathrm{Map}\left(\mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}^{\Delta^n}\right)_{/x}, \mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}^{\Delta^m}\right)_{/y}\right)$$

as the adjoint to the map

$$\mathrm{Hom}_{\mathbf{\Delta}}\left([m], [n]\right) \times \mathcal{C}^{\Delta^m}\left(x, y\right) \times \mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}^{\Delta^n}\right)_{/x} \to \mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}^{\Delta^m}\right)_{/y}$$

which we may describe as follows: A k-simplex on the left hand side is a triple (u, f, g), where $u : [m] \to [n]$ is a poset map, $f \in \mathcal{C}^{\Delta^m}\left(x, y\right)_k$, and $g\colon \mathfrak{C}[\Delta^{k+1}] \to \mathcal{C}^{\Delta^n}$ is a simplicial functor carrying $k+1$ to x. We declare that its image is the k-simplex of $\mathrm{N}_{\mathrm{hc}}\left(\mathcal{C}^{\Delta^m}\right)_{/y}$ corresponding to the simplicial functor $h\colon \mathfrak{C}[\Delta^{k+1}] \to \mathcal{C}^{\Delta^m}$ defined by

$$h\left(i\right) = \begin{cases} g\left(i\right) & \text{if } 0 \leq i \leq k, \\ y & \text{if } i = k+1. \end{cases}$$

For $0 \leq i \leq j \leq k$, the map $h_{i,j}\colon \mathfrak{C}[\Delta^{k+1}]\left(i, j\right) \to \mathcal{C}^{\Delta^m}\left(h\left(i\right), h\left(j\right)\right)$ is the composite

$$\mathfrak{C}[\Delta^{k+1}]\left(i, j\right) \xrightarrow{g} \mathcal{C}^{\Delta^n}\left(g\left(i\right), g\left(j\right)\right) \xrightarrow{u^*} \mathcal{C}^{\Delta^m}\left(g\left(i\right), g\left(j\right)\right).$$

For $0 \leq i \leq k$, the map $h_{0,k+1}\colon \mathfrak{C}[\Delta^{k+1}]\left(i, k+1\right) \to \mathcal{C}^{\Delta^m}\left(h\left(i\right), k+1\right)$ is the composite

$$\mathfrak{C}[\Delta^{k+1}]\left(i, k+1\right) \xrightarrow{(\max, g)} \Delta^k \times \mathcal{C}^{\Delta^n}\left(g\left(i\right), x\right) \xrightarrow{f \times u^*} \mathcal{C}^{\Delta^m}\left(x, y\right) \times \mathcal{C}^{\Delta^m}\left(g\left(i\right), x\right) \xrightarrow{\circ} \mathcal{C}^{\Delta^m}\left(g\left(i\right), y\right),$$

where $\max\colon \mathfrak{C}[\Delta^{k+1}]\left(i, k+1\right) \to \Delta^k$ is the nerve of the poset map

$$\max : \left(\{S \subset [k] \mid \min S = \{i\}\}, \subset\right) \to [k].$$

□

3. Decomposing the G-action on $\mathbb{E}_M$

Let M be a manifold with an action of a topological group G. As we saw in Remark 1.6, the ∞-operads $|\mathrm{Disk}\,(M)| \simeq \mathbb{E}_M$ inherits a G-action from M. The goal of this section is to "decompose" this action into pieces. More precisely, we are going to construct a functor

$$\overline{\mathrm{Disk}}\,(M)_\bullet : \mathrm{Flat}\,(BG) \to \mathsf{Op}$$

such that $\left|\overline{\mathrm{Disk}}\,(M)_\bullet\right|$ carries all maps to equivalences. We will then show that the induced functor $BG \simeq |\mathrm{Flat}\,(BG)| \to \mathcal{O}\mathsf{p}_\infty$ is equivalent to the G-action on $\mathbb{E}_M$ (Theorem 3.2).

We begin with the definition of $\overline{\mathrm{Disk}}\,(M)_\bullet$.

Construction 3.1. Let M be a d-manifold. For each $n \geq 0$, we define an operad $\overline{\mathrm{Disk}}\,(M)_n$ as follows:

(1) Objects are images of continuous maps $\phi\colon \Delta^n \times \mathbb{R}^d \to M$ such that the map
$$\Delta^n \times \mathbb{R}^d \to \Delta^n \times M,\ (t,x) \mapsto (t, \phi\,(t,x))$$
is an open embedding (which is automatic if ϕ is injective [Lur17, Theorem 5.4.1.1]). Note that such an object is completely determined by a family of open sets $U_\bullet = (U_t)_{t\in\Delta^n}$.

(2) There is a unique map $(U_{i,\bullet})_{i\in I} \to V_\bullet$ if for each $t \in \Delta^n$, the sets $(U_{i,t})_{i\in I}$ are disjoint and contained in V_t. There is no such morphism otherwise.

Suppose now that M admits a left action by a topological group G. In this situation, the operad $\overline{\mathrm{Disk}}\,(M)_n$ carries an action of $G_n = (\mathrm{Sing}\,G)_n$: For each map $g\colon \Delta^n \to G$, the map $g_*\colon \overline{\mathrm{Disk}}\,(M)_n \to \overline{\mathrm{Disk}}\,(M)_n$ carries an object $U_\bullet$ to $g \cdot U_\bullet = (g\,(t) \cdot U_t)_{t\in\Delta^n}$. Moreover, this action is compatible with $[n] \in \mathbf{\Delta}^{\mathrm{op}}$, in the sense that the diagram

$$\begin{tikzcd}
\overline{\mathrm{Disk}}(M)_n \arrow[r, "u^*"] \arrow[d, "g_*"'] & \overline{\mathrm{Disk}}(M)_m \arrow[d, "(u^*g)_*"] \\
\overline{\mathrm{Disk}}(M)_n \arrow[r, "u^*"'] & \overline{\mathrm{Disk}}(M)_m
\end{tikzcd}$$

commutes for every poset map $u\colon [m] \to [n]$ and $g \in G_n$. Thus, we obtain a functor

$$\mathrm{Flat}\,(BG) = \int^{[n]\in\mathbf{\Delta}^{\mathrm{op}}} BG_n \to \mathsf{Op} \xrightarrow{|-|} \mathcal{O}\mathsf{p}_\infty, [n] \mapsto \left|\overline{\mathrm{Disk}}\,(M)_n\right|.$$

We can now state the main result of this section.

Theorem 3.2. *Let M be a topological manifold equipped with a left action of a topological group G.*

(1) The functor
$$\left|\overline{\mathrm{Disk}}\,(M)_\bullet\right| : \mathrm{Flat}\,(BG) \to \mathcal{O}\mathsf{p}_\infty$$
carries all morphisms to equivalences.

(2) The induced functor
$$BG \simeq |\mathrm{Flat}\,(BG)| \to \mathcal{O}\mathsf{p}_\infty$$
is naturally equivalent to $\mathbb{E}_M\colon BG \to \mathcal{O}\mathsf{p}_\infty$.

The remainder of this section is devoted to the proof of Theorem 3.2. We need a few preliminaries.

Proposition 3.3. *Let M be a d-manifold. For every $n \geq 0$, the map $\left|\overline{\mathrm{Disk}}\,(M)_0\right| \to \left|\overline{\mathrm{Disk}}\,(M)_n\right|$ is an equivalence of ∞-operads.*

Proof. We will use the fiberwise criterion. We must check the following:

(1) The map $U\left(\overline{\mathrm{Disk}}\left(M\right)_0\right) \to U\left(\overline{\mathrm{Disk}}\left(M\right)_n\right)$ is a weak homotopy equivalence.

(2) For every $k \geq 0$ and every $V \in \overline{\mathrm{Disk}}\left(M\right)_0$, the map

$$\mathrm{Alg}_{C_k}\left(\overline{\mathrm{Disk}}\left(M\right)_0\right)_V \to \mathrm{Alg}_{C_k}\left(\overline{\mathrm{Disk}}\left(M\right)_n\right)_{(V)_{t\in\Delta^n}}$$

is a weak homotopy equivalence.

(3) For every $k \geq 0$ and every morphism $V_\bullet \to V'_\bullet$ in $\overline{\mathrm{Disk}}\left(M\right)_n$, the map

$$\mathrm{Alg}_{C_k}\left(\overline{\mathrm{Disk}}\left(M\right)_n\right)_{V_\bullet} \to \mathrm{Alg}_{C_k}\left(\overline{\mathrm{Disk}}\left(M\right)_n\right)_{V'_\bullet}$$

is a weak homotopy equivalence.

For part (1), consider the functor

$$\Phi\colon U\left(\overline{\mathrm{Disk}}\left(M\right)_n\right) \to \mathrm{Open}\left(\mathrm{Map}\left(\Delta^n, M\right)\right)$$

which carries an object $U_\bullet$ to the contractible open set

$$\Phi\left(U_\bullet\right) = \{f \in \mathrm{Map}\left(\Delta^n, M\right) \mid f\left(t\right) \in U_t \text{ for all } t \in \Delta^n\}.$$

We remark that $\Phi\left(U_\bullet\right)$ is contractible, because it is homeomorphic to $\mathrm{Map}\left(\Delta^n, \mathbb{R}^d\right)$. It follows from Seifert–van Kampen's theorem and Lemmas A.1 and A.2 that

$$\left|U\left(\overline{\mathrm{Disk}}\left(M\right)_n\right)\right| \simeq \mathrm{colim}_{U_\bullet \in U(\mathrm{Disk}^n(M))}\, \Phi\left(U_\bullet\right) \simeq \mathrm{Map}\left(\Delta^n, M\right).$$

Thus, we are reduced to showing that the map $\mathrm{Map}\left(\Delta^0, M\right) \to \mathrm{Map}\left(\Delta^n, M\right)$ is a homotopy equivalence, which is obvious.

Part (2) follows by a similar argument, noting that the relevant map is identified with $\mathrm{Map}\left(\Delta^0, \mathrm{Conf}_k\left(\mathbb{R}^d\right)\right) \to \mathrm{Map}\left(\Delta^n, \mathrm{Conf}_k\left(\mathbb{R}^d\right)\right)$.

For (3), write $E\left(V_\bullet\right) \subset \mathrm{Map}\left(\Delta^n, \mathrm{Conf}_k\left(M\right)\right)$ for the subspace consisting of the functions $f\colon \Delta^n \to \mathrm{Conf}_k\left(M\right)$ such that $f\left(t\right) \in \mathrm{Conf}_k\left(V_t\right)$ for every $t \in \Delta^n$. Arguing exactly as above, we are reduced to showing that the inclusion

$$E\left(V_\bullet\right) \hookrightarrow E\left(V'_\bullet\right)$$

is a homotopy equivalence. One can identify $E\left(V_\bullet\right)$ and $E\left(V'_\bullet\right)$ with $\mathrm{Map}\left(\Delta^n, \mathrm{Conf}_k\left(\mathbb{R}^d\right)\right)$, so it will suffice to prove the following: For any continuous map $\phi\colon \Delta^n \to \mathrm{Emb}\left(\mathbb{R}^d, \mathbb{R}^d\right)$, the induced map

$$\begin{aligned}\mathrm{Map}\left(\Delta^n, \mathrm{Conf}_k\left(\mathbb{R}^d\right)\right) &\to \mathrm{Map}\left(\Delta^n, \mathrm{Conf}_k\left(\mathbb{R}^d\right)\right),\\ \left(f_1, \ldots, f_k\right) &\mapsto \left(t \mapsto \left(\phi\left(t\right) \circ f_1\left(t\right), \ldots, \phi\left(t\right) \circ f_k\left(t\right)\right)\right)\end{aligned}$$

is a homotopy equivalence. This follows from Kister–Mazur's theorem, which says that ψ is homotopic to a homeomorphism. The claim follows. □

For later reference, we record claim (3) in the above proof as a separate lemma.

Lemma 3.4. *Let M be a d-manifold. For every $k \geq 0$ and every morphism $V_\bullet \to V'_\bullet$ in $\overline{\mathrm{Disk}}\left(M\right)_n$, the map*

$$\mathrm{Alg}_{C_k}\left(\overline{\mathrm{Disk}}\left(M\right)_n\right)_{V_\bullet} \to \mathrm{Alg}_{C_k}\left(\overline{\mathrm{Disk}}\left(M\right)_n\right)_{V'_\bullet}$$

is a weak homotopy equivalence.

Going back to the proof of Theorem 3.2, note that Proposition 3.3 addresses part (1) of Theorem 3.2. To address part (2), we introduce an ∞-operadic version of $\overline{\mathrm{Disk}}\left(M\right)_n$, denoted by $\mathbb{E}_{M,(n)}$.

Definition 3.5. Let $n, d \geq 0$. We define an ∞-category $\mathcal{M}\mathsf{fld}\left(d\right)_{(n)}$ as the homotopy coherent nerve of the simplicial category $\mathcal{M}\mathsf{fld}\left(d\right)^{\mathbf{\Delta},\Delta^n}$ (Definitions 1.1 and 2.1). We write $B\,\mathrm{Emb}\left(d\right)_{(n)} \subset \mathcal{M}\mathsf{fld}\left(d\right)_{(n)}$ for the full subcategory spanned by $\mathbb{R}^d$.

Given a Kan fibration $p\colon B \to B\,\mathrm{Emb}\,(d)_{(n)}$, we define an ∞-operad $\mathbb{E}_{B,(n)}$ as the suboperad of $B^{\amalg}$ consisting of those operations $(\alpha_i)_{i\in I}:\ (b_i)_{i\in I} \to b$ with the following property: For each $t \in \Delta^n$, the restriction of the maps $\{p\,(\alpha_i) : \Delta^n \times \mathbb{R}^d \to \mathbb{R}^d\}_{i\in I}$ to $\{t\} \times \mathbb{R}^d$ have disjoint images.

Given a d-manifold M, we will write $\mathbb{E}_{M,(n)} = \mathbb{E}_{B\,\mathrm{Emb}(d)_{(n)/M}}$.

Proof of Theorem 3.2. As we already mentioned, part (1) follows from Proposition 3.3. For part (2), we observe that the maps

$$\overline{\mathrm{Disk}}\,(M)_n \to \mathbb{E}_{M,(n)}$$

determines a natural transformation $\mathrm{Disk}\,(M)_\bullet \Rightarrow \mathbb{E}_{M,(\bullet)}\colon \mathrm{Flat}\,(BG) \to \mathcal{O}\mathsf{p}_\infty$. According to [Ram22, Proposition A.11], this natural transformation descends to a natural transformation $|\mathrm{Disk}\,(M)_\bullet| \Rightarrow \mathbb{E}_{M,(\bullet)}$, which is a natural equivalence by Proposition 3.3 and Theorem 1.8. Thus, we are reduced to showing that the functor

$$\mathbb{E}_{M,(\bullet)}\colon BG \simeq |\mathrm{Flat}\,(BG)| \to \mathcal{O}\mathsf{p}_\infty$$

is equivalent to the functor $\mathbb{E}_M\colon BG \to \mathcal{O}\mathsf{p}_\infty$. This follows from Proposition 2.7, and the proof is complete. □

4. Analyzing pieces

Let M be a topological manifold with a left action of a topological group G. In the previous section, we decomposed the G-action on $\mathbb{E}_M$ into pieces $\{\overline{\mathrm{Disk}}\,(M)_n\}_{n\geq 0}$, where each $\overline{\mathrm{Disk}}\,(M)_n$ admits a G_n-action. The operad $\overline{\mathrm{Disk}}\,(M)_n \ltimes G_n$ looks very much like $\mathrm{Disk}_G\,(M)_n$, except that it is slightly bigger. More precisely, we can identify $\mathrm{Disk}_G\,(M)_n$ as a full suboperad of $\overline{\mathrm{Disk}}\,(M)_n \ltimes G_n$ spanned by the constant objects, i.e., objects $U_\bullet$ such that U_t does not depend on $t \in \Delta^n$. The goal of this section is to show this enlargement of $\mathrm{Disk}_G\,(M)_n$ does not affect locally constant algebras over them. More precisely, we prove the following:

Theorem 4.1. *Let M be a topological manifold with a left action of a topological group G. For each $n \geq 0$, the inclusion*

$$\mathrm{Disk}_G\,(M)_n \hookrightarrow \overline{\mathrm{Disk}}\,(M)_n \ltimes G_n,\ U \mapsto (U)_{t\in\Delta^n}$$

induces an equivalence of ∞-operads

$$|\mathrm{Disk}_G\,(M)_n| \xrightarrow{\simeq} \left|\overline{\mathrm{Disk}}\,(M)_n \ltimes G_n\right|.$$

The proof of Theorem 4.1 relies on the following lemma.

Lemma 4.2. *Let $\mathcal{O}$ be a discrete operad with an action of a discrete group Γ, and let $k \geq 0$. Suppose that for every unary morphism $y \to y'$, the map*

$$\mathrm{Alg}_{C_k}\,(\mathcal{O})_y \to \mathrm{Alg}_{C_k}\,(\mathcal{O})_{y'}$$

is a weak homotopy equivalence. Then $\mathcal{O} \ltimes \Gamma$ has the same property.

Proof. Both the source and the target projects to the groupoid $B\Gamma^k$ via isofibrations (hence cocartesian fibrations). Therefore, by Quillen's Theorem B, it suffices to show that the map between the (strict) fibers of these projections is a weak homotopy equivalence. But this map can be identified with the map

$$\coprod_{g\in\Gamma} \mathrm{Alg}_{C_k}\,(\mathcal{O})_y \to \coprod_{g\in\Gamma} \mathrm{Alg}_{C_k}\,(\mathcal{O})_y\,,$$

which is a weak homotopy equivalence by hypothesis. The claim follows. □

Proof of Proposition 4.1. Throughout the proof, we will write $\mathcal{O} = \mathrm{Disk}_G\,(M)_n$ and $\mathcal{P} = \overline{\mathrm{Disk}}\,(M)_n \ltimes G_n$.

We use the fiberwise criterion. We must verify the following conditions:

(1) The functor $U(\mathcal{O}) \to U(\mathcal{P})$ is a weak homotopy equivalence.
(2) For every $k \geq 0$ and every $V \in \mathcal{O}$, the map
$$\mathrm{Alg}_{C_k}(\mathcal{O})_V \to \mathrm{Alg}_{C_k}(\mathcal{P})_{(V)_{t\in\Delta^n}}$$
is a weak homotopy equivalence.
(3) For every $k \geq 0$ and every morphism $V_\bullet \to V'_\bullet$ in $\mathcal{P}$, the map
$$\mathrm{Alg}_{C_k}(\mathcal{P})_{V_\bullet} \to \mathrm{Alg}_{C_k}(\mathcal{P})_{V'_\bullet}$$
is a weak homotopy equivalence.

We start with (1). Set $EG_n = (BG_n)_{*/}$. By Quillen's Theorem B, it suffices to show that the functor
$$EG_n \times_{BG_n} U(\mathcal{O}) \to EG_n \times_{BG_n} U(\mathcal{P})$$
is a weak homotopy equivalence. We will prove this by identifying the homotopy type of the source and target.

We first identify the weak homotopy type of the target. Unwinding the definitions, the category $EG_n \times_{BG_n} U(\mathcal{P})$ admits the following description:

- Its objects are the pairs $(g_\bullet, U_\bullet)$, where $g\colon \Delta^n \to G$ is a continuous map and $U_\bullet \in \mathcal{P}$.
- There is a unique morphism $(g_\bullet, U_\bullet) \to (g'_\bullet, U'_\bullet)$ if $g_t U_t \subset g'_t U'_t$ for all $t \in \Delta^n$. Otherwise, there is no morphism.

For each $(g_\bullet, U_\bullet) \in EG_n \times_{BG_n} U(\mathcal{P})$, let $\Phi(g_\bullet, U_\bullet)$ denote the (contractible) open set of $\mathrm{Map}(\Delta^n, M)$ consisting of those functions $f\colon \Delta^n \to M$ such that $f(t) \in g_t U_t$ for all $t \in \Delta^n$. Using Seifert–van Kampen's theorem and Lemmas A.1 and A.2, we compute
$$|EG_n \times_{BG_n} U(\mathcal{P})| \simeq \mathrm{colim}_{(g_\bullet,U_\bullet)\in EG_n\times_{BG_n}U(\mathcal{P})} \Phi(g_\bullet, U_\bullet) \simeq \mathrm{Map}(\Delta^n, M).$$

Next, we identify the weak homotopy type of $EG_n \times_{BG_n} U(\mathcal{O})$. This category admits a similar description as $EG_n \times_{BG_n} U(\mathcal{P})$, except that objects are pairs $(g_\bullet, U)$ with $U \in \mathcal{O}$. Arguing exactly as above, we obtain
$$|EG_n \times_{BG_n} U(\mathcal{P})| \simeq \bigcup_{(g_\bullet,U)\in EG_n\times_{BG_n}U(\mathcal{O})} \Phi\left(g_\bullet, (U)_{t\in\Delta^n}\right),$$
where the right hand side is the (set-theoretic) union of the open subspaces of $\mathrm{Map}(\Delta^n, M)$, equipped with the subspace topology.

In light of the previous two paragraphs, the proof of (1) reduces to showing that the inclusion
$$\bigcup_{(g_\bullet,U)\in EG_n\times_{BG_n}U(\mathcal{O})} \Phi\left(g_\bullet, (U)_{t\in\Delta^n}\right) \hookrightarrow \mathrm{Map}(\Delta^n, M)$$
is a homotopy equivalence. For this, it suffices to show that the space $X = \bigcup_{(g_\bullet,U)\in EG_n\times_{BG_n}U(\mathcal{O})} \Phi\left(g_\bullet, (U)_{t\in\Delta^n}\right)$ deformation retracts onto the subspace of constant maps $\Delta^n \to M$. This is indeed the case: A deformation retraction is given by
$$[0,1] \times X \to X,\ (s, f) \mapsto (t \mapsto f(st_0 + (1-s)t)),$$
where $t_0 \in \Delta^n$ is any of the reader's favorite point.

The proof of (2) proceeds in an entirely similar manner. It will suffice to show that the map
$$EG_n^k \times_{BG_n^k} \mathrm{Alg}_{C_k}(\mathcal{O})_V \to EG_n^k \times_{BG_n^k} \mathrm{Alg}_{C_k}(\mathcal{P})_{(V)_{t\in\Delta^n}}$$
is a homotopy equivalence. Arguing exactly as above, we find that both the source and target has the weak homotopy type of $\mathrm{Map}(\Delta^n, \mathrm{Conf}_k(V))$.

Part (3) is a consequence of Lemmas 3.4 and 4.2 (applied to the G_n-action on $\mathcal{P}$). The proof is now complete. □

5. Proof of the main theorem

We now have all the necessary ingredients to prove Theorem 0.2.

Proof of Theorem 0.2. We consider the following chain of equivalences:

$$\begin{aligned}
|\mathrm{Disk}_G(M)| &\underset{(1)}{\simeq} \left|\mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathrm{Disk}_G(M)_n\right| \\
&\underset{(2)}{\simeq} \left|\mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} |\mathrm{Disk}_G(M)_n|\right| \\
&\underset{(3)}{\simeq} \left|\mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \left|\overline{\mathrm{Disk}}(M)_n \ltimes G_n\right|\right| \\
&\underset{(4)}{\simeq} \left|\mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \overline{\mathrm{Disk}}(M)_n \ltimes G_n\right| \\
&\underset{(5)}{\simeq} \left|\mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathrm{colim}_{BG_n} \overline{\mathrm{Disk}}(M)_n\right| \\
&\underset{(6)}{\simeq} \left|\mathrm{colim}_{[n]\in\int BG_\bullet} \overline{\mathrm{Disk}}(M)_n\right| \\
&\underset{(7)}{\simeq} \left|\mathrm{colim}_{[n]\in\int BG_\bullet} \left|\overline{\mathrm{Disk}}(M)_n\right|\right| \\
&\underset{(8)}{\simeq} \left|\mathrm{colim}_{BG} \mathbb{E}_M\right| \\
&\underset{(9)}{\simeq} \mathrm{colim}_{BG} \mathbb{E}_M.
\end{aligned}$$

Each of the equivalences is justified as follows.

- (1) is an instance of Corollary 2.4.
- (2) is an instance of Remark 0.8.
- (3) follows from Theorem 4.1.
- (4) is another instance of Remark 0.8.
- (5) follows from the fact that localization of operadic Grothendieck construction at cocartesian morphsims model colimits. (In this case, all cocartesian maps are already equivalences because BG_n is a groupoid.)
- (6) follows from a general result on left Kan extensions along cocartesian fibrations [Lur25, Tag 02ZM].
- (7) is another instance of Remark 0.8.
- (8) follows from Theorem 3.2 (and the fact that localizations are final [Cis19b, Proposition 7.1.10].
- (9) follows from the observation that $U(\mathrm{colim}_{BG} \mathbb{E}_M)$ is already a groupoid, which in turn follows from the description of colimits of ∞-operads in terms of operadic Grothendieck construction.

□

Appendix A. Miscellaneous results

This section collects miscellaneous results on manifolds that we use in the main body of the paper.

Lemma A.1. *Let M be a d-manifold. The evaluation at the origin*

$$\varepsilon\colon \mathrm{Emb}\left(\mathbb{R}^d \times \{1,\dots,k\}, M\right) \to \mathrm{Conf}_k(M)$$

has a right lifting property for the inclusion $\emptyset \hookrightarrow \Delta^n$

Proof. The claim is obvious if $n = 0$, so it suffices to show that ε has the right lifting property for the inclusion $\Delta^0 \hookrightarrow \Delta^n$. This follows from [Ara24a, Proposition C.7], which asserts that ε is a Serre fibration. □

Lemma A.2. *Let M be a d-manifold, and let $\phi, \psi \colon \Delta^n \times \mathbb{R}^d \to M$ be maps such that ϕ_t and ψ_t are embeddings for each $t \in \Delta^n$. If there is a map $f \colon \Delta^n \to M$ such that $f(t) \in \phi_t(\mathbb{R}^d) \cap \psi_t(\mathbb{R}^d)$ for all $t \in \Delta^n$, then there is some $r > 0$ that satisfies*

$$f(t) \in \phi_t(B_d(r)) \subset \psi_t(\mathbb{R}^d)$$

for all $t \in \Delta^n$, where $B^d(r) = \{x \in \mathbb{R}^d \mid |x| < r\}$.

Proof. By hypothesis, the map $(\mathrm{id}, f) : \Delta^n \to \Delta^n \times M$ can be factored as

$$\Delta^n \xrightarrow{(\mathrm{id}, f')} \Delta^n \times \mathbb{R}^d \xrightarrow{\phi} \Delta^n \times M.$$

Consider the set $U \subset \Delta^n \times \mathbb{R}^d$ of points (t, x) such that $\phi_t(x) \in \psi_t(\mathbb{R}^n)$. This is an open set by [Lur17, Theorem 5.4.1.1], and (id, f') factors through U by hypothesis. Using the compactness of Δ^n, we can find some $r > 0$ with the following property: For each $t \in \Delta^n$ and y, we have $(t, f'(t) + y) \in U$. This r has the desired property. □

References

[Ara24a] Kensuke Arakawa, *A context for manifold calculus*, `https://arxiv.org/abs/2403.03321`, 2024, To appear in *J. Homotopy Relat. Struct.*

[Ara24b] ———, *The Grothendieck construction for ∞-categories fibered over categorical patterns*, `https://arxiv.org/abs/2404.01025`, 2024.

[Ara25] Kensuke Arakawa, *Universal properties of variations of the little cubes operads*, Münster J. Math. **18** (2025), no. 1, 1–26. MR 4976171

[Cis19a] Denis-Charles Cisinski, *Higher categories and homotopical algebra*, Cambridge Studies in Advanced Mathematics, vol. 180, Cambridge University Press, Cambridge, 2019. MR 3931682

[Cis19b] ———, *Higher categories and homotopical algebra*, Cambridge Studies in Advanced Mathematics, vol. 180, Cambridge University Press, Cambridge, 2019. MR 3931682

[DK85] W. G. Dwyer and D. M. Kan, *Equivariant homotopy classification*, J. Pure Appl. Algebra **35** (1985), no. 3, 269–285. MR 777259

[DK87] ———, *Equivalences between homotopy theories of diagrams*, Algebraic topology and algebraic K-theory (Princeton, N.J., 1983), Ann. of Math. Stud., vol. 113, Princeton Univ. Press, Princeton, NJ, 1987, pp. 180–205. MR 921478

[HM22] Gijs Heuts and Ieke Moerdijk, *Simplicial and dendroidal homotopy theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. A Series of Modern Surveys in Mathematics [Results in Mathematics and Related Areas. 3rd Series. A Series of Modern Surveys in Mathematics], vol. 75, Springer, Cham, [2022] ©2022. MR 4485749

[Kis64] J. M. Kister, *Microbundles are fibre bundles*, Ann. of Math. (2) **80** (1964), 190–199. MR 180986

[Lur09] Jacob Lurie, *Higher topos theory*, Annals of Mathematics Studies, vol. 170, Princeton University Press, Princeton, NJ, 2009. MR 2522659

[Lur17] J. Lurie, *Higher algebra*, `https://www.math.ias.edu/~lurie/papers/HA.pdf`, 2017.

[Lur25] Jacob Lurie, *Kerodon*, `https://kerodon.net`, 2025.

[Ram22] Maxime Ramzi, *A monoidal grothendieck construction for ∞-categories*, `https://arxiv.org/abs/2209.12569`, 2022.

[WY24] David White and Donald Yau, *Smith ideals of operadic algebras in monoidal model categories*, Algebr. Geom. Topol. **24** (2024), no. 1, 341–392. MR 4721371